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Beyond Isolated Words: Diffusion Brush for Handwritten Text-Line Generation

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Abstract

Existing handwritten text generation methods primarily focus on isolated words. However, realistic handwritten text demands attention not only to individual words but also to the relationships between them, such as vertical alignment and horizontal spacing. Therefore, generating entire text line emerges as a more promising and comprehensive task. However, this task poses significant challenges, including the accurate modeling of complex style patterns—encompassing both intra- and inter-word relationships—and maintaining content accuracy across numerous characters. To address these challenges, we propose DiffBrush, a novel diffusion-based model for handwritten text-line generation. Unlike existing methods, DiffBrush excels in both style imitation and content accuracy through two key strategies: (1) content-decoupled style learning, which disentangles style from content to better capture intra-word and inter-word style patterns by using column- and row-wise masking; and (2) multi-scale content learning, which employs line and word discriminators to ensure global coherence and local accuracy of textual content. Extensive experiments show that DiffBrush excels in generating high-quality text-lines, particularly in style reproduction and content preservation. Code is available at https://github.com/dailenson/DiffBrush

1. Introduction

Handwritten text generation aims to automatically synthesize realistic handwritten text images that visually convey a user's personal writing style (*e.g.*, text slant, stroke width, ligatures) while ensuring the content readability. This task has broad applications, including assisting individuals with writing difficulties, accelerating handwritten font design, and enriching data for text recognizer. Most existing methods [1, 3, 7, 11, 12, 35, 44] focus on generating handwritten images at the word level, with few efforts [9, 23] exploring the generation of complete text lines. To bridge this gap, our work focuses on high-quality handwritten text-line generation with better control over both style and content.

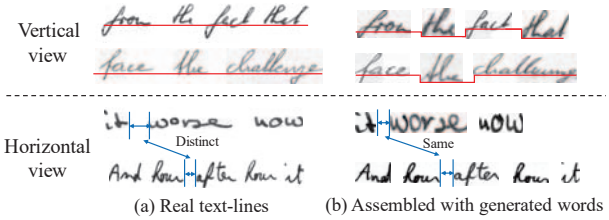
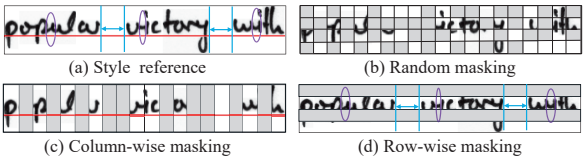


Figure 1. Comparison of handwritten text-lines (a) written by real writers, and (b) assembled with isolated words generated from One-DM [7]. The latter one applies fixed inter-word spacing due to the lack of spacing information in generated words. Red lines indicate the baseline (*i.e.*, the reference line at the bottom of the characters), while blue lines highlight word spacing.

Most previous state-of-the-art methods [7, 12, 40, 44] focus on handwritten word generation by using reference images from writers as style inputs and conditioning on character-wise labels or images for content. This allows for the synthesis of handwritten words with controllable styles and specific content. However, as shown in Figure 1, generating text at the word level cannot effectively capture the cohesive style of a complete text line: (1) Humans generally maintain consistent vertical alignment across words, while synthesized words often exhibit arbitrary vertical positioning. (2) Different writers have unique word spacing characteristics that are often lost in isolated word generation.

Direct approaches for text-line generation are relatively limited, with two notable GAN-based methods proposed. TS-GAN [9] optimizes a global content recognition loss based on the entire generated text-line image, primarily guiding content learning while implicitly influencing style learning. CSA-GAN [23], in contrast, leverages both a content recognition loss and a writer classification loss computed from the generated text-line image, thus better modeling style through writer identity supervision. However, both methods suffer from two key limitations: 1) **Ineffective style extraction**: Since both methods jointly optimize content and style from the same model output, the two aspects interfere with each other, preventing effective learn-

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(a) Style reference (b) Random masking (c) Column-wise masking (d) Row-wise masking

Figure 2. (a) This style reference exhibits rich style patterns, *e.g.*, ligatures, spacing, and vertical alignment alongside undesired content information: "popular victory with". (b) Random masking disrupts both style features and content information. (c) Column-wise masking maintains style patterns (*e.g.,* character style and vertical alignment) while removing horizontal content information. (d) Row-wise masking preserves joins and spacing while disrupting vertical content.

ing of either. For instance, minimizing content recognition loss inevitably pushes these models to produce easily recognizable outputs with simplified styles (*e.g.*, regular fonts and standard strokes, as shown in Figure 9), ultimately hindering the faithful mimicking of diverse handwriting styles. 2) **Difficulty in maintaining character-level accuracy**: Ensuring the readability of text lines with numerous characters remains challenging. For instance, in datasets like IAM [36], where a single text line averages 42 characters—approximately six times the length of a typical word. Optimizing content loss at the text-line level encourages global correctness but may fail to preserve individual character accuracy, making it difficult to maintain content integrity across the entire generated text (cf. D_{CER} and D_{WER} columns in Table 1).

To generate handwritten text lines with improved content accuracy and style fidelity, we introduce **DiffBrush**, a novel diffusion-based approach. Our method incorporates two key strategies: (1) **content-decoupled style learning**, and (2) **multi-scale content learning**.

Content-decoupled style learning aims to disrupt content information of style references while preserving key style patterns. This eliminates content interference, thus achieving effective one-shot style learning (cf. Table 1 and Figure 6). A naive approach, such as random masking, fails as it disrupts both content and key style features like word spacing and vertical alignment (cf. Figure 2(b)). To address this, we propose column- and row-wise masking (cf. Figure 2(c), (d)), which selectively disrupts content while preserving critical style patterns. For style enhancement in vertical direction, as shown in Figure 3, we apply column-wise masking to the extracted style features, maintaining vertical alignment while removing horizontal content information. A Proxy-NCA loss [25, 38] is then used to enforce style consistency within writers while distinguishing different writers, enabling the vertical enhancing head to refine vertical alignment. For style enhancement in horizontal direction, row-wise masking preserves word and character spacing while disrupting vertical content, allowing the horizontal enhancing head to reinforce horizontal spacing patterns. This novel masking strategy effectively disentangles style from content, enabling more accurate and independent style representation in handwritten text-line generation.

Multi-scale content learning seeks to enhance content accuracy at both global and local levels: at the global level, we preserve character order within a text line to maintain contextual relationships between characters, while at the local level, we ensure the structural correctness of each individual word. To achieve this, we develop a novel multi-scale content discriminator. The line content discriminator segments the text-line image and processes it with a 3D CNN [55] to capture global contextual relationships, encouraging the generator to maintain proper character sequencing. Meanwhile, the word discriminator employs an attention mechanism to isolate individual words and verify their content accuracy, guiding the generator to refine local text content. Our empirical results (cf. Figure 7) show that this multi-scale content discriminator significantly improves content accuracy without hindering style imitation quality.

Our main contributions include: (1) To the best of our knowledge, DiffBrush is among the first to leverage diffusion generative models for handwritten text-line generation. (2) DiffBrush introduces a novel content-decoupled style learning strategy that significantly enhances style imitation, along with a new multi-scale content learning strategy that boosts content accuracy. (3) Extensive experiments on two popular English handwritten datasets (cf. Table 1 and Figure 6) and one Chinese dataset (cf. Figure 11) demonstrate that DiffBrush significantly outperforms state-of-the-arts.

2. Related Work

Handwritten text generation methods are generally divided into online and offline: the former synthesizes dynamic stroke sequences, while the latter generates static text images. With the advancement of deep learning, Transformer decoders [6] and diffusion models [8, 34, 47] have been used for synthesizing online handwritten text. However, as highlighted in recent studies [3, 7, 44], online methods require temporal data (*e.g.*, coordinate points and writing orders) collected from a digital stylus pen and cannot synthesize stroke width, ink color like offline methods. In light of this, this paper focuses on offline handwriting generation.

The advent of Generative Adversarial Networks [20, 33] has accelerated the development of offline handwritten text generation. Early works [1, 11] use character labels as content inputs and random noise as style inputs to synthesize handwritten words with controllable content and random styles. To enhance style control, SLOGAN [35] conditions style inputs on fixed writer IDs but fails to mimic unseen styles. Unlike them, GANwriting [22] and HWT [3] employ CNN or transformer encoder to extract style features from style references and are thus capable of imitating any styles. Further, VATr [44] utilizes symbol images as content representations, enabling character generation beyond

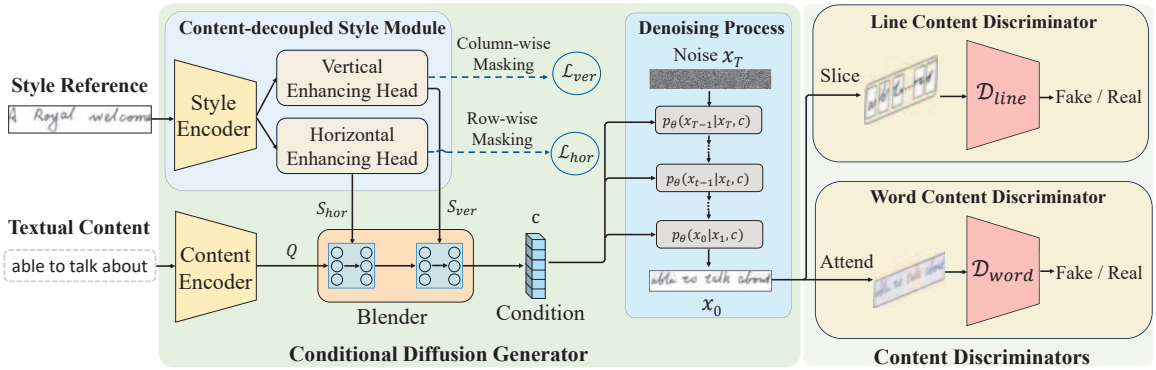


Figure 3. Method overview. Our DiffBrush consists of a conditional diffusion generator, a content-decoupled style module, and multi-scale content discriminators. The style module extracts two enhanced style features S_{hor} and S_{ver}, which are blended with the content representation Q from the content encoder to construct the condition vector c. This condition is then used to guide the denoising process to generate images. To enhance style learning, we explore column- and row-wise masking to effectively eliminate content interference in style modeling. The content discriminators provide feedback at both line and word levels, enhancing the readability of generated text. The slice operation divides the text-line image into horizontal segments, while the attend operation locates word positions in the text-line.

the training charset. In contrast to the above word-focused methods, TS-GAN [9] and CSA-GAN [23] are developed to synthesize handwritten text-lines. However, they struggle to produce satisfactory results due to design drawbacks in style learning and content supervision.

The rapid development of diffusion models [10, 19, 28, 58, 60] offers new potential for handwritten text generation. However, some early attempts [39, 67], which condition denoising process on fixed writer labels, cannot mimic unseen handwriting styles. To address this, DiffusionPen [40] and One-DM [7] extract style information from reference images, and then merge this information with the textual content to guide the denoising process. However, regarding text content readability, One-DM simply incorporates a text recognizer with a CTC decoder, while DiffusionPen neglects this challenge entirely. Different from them, we propose a novel multi-scale content learning strategy, significantly enhancing text content readability. Moreover, previous diffusion methods [7, 39, 40, 67] focus on generating isolated words whereas our DiffBrush aims for entire text-line generation. We discuss more related works about diffusion methods for general image generation in Appendix A.

3. Problem Statement and Preliminaries

Problem statement. Given a text string $\mathcal{A}$ and a style reference s_i randomly sampled from an exemplar writer $w_i \in \mathcal{W}$, we aim to synthesize a handwritten text-line image x that captures the unique calligraphic style of w_i while accurately preserving the content of $\mathcal{A}$. Here, $\mathcal{A} = \{a_i\}_{i=1}^{L}$ represents a sequence of length L, where each a_i is a Unicode character, including lowercase and uppercase letters, digits, punctuation. The key challenges lie in accurately capturing handwriting styles, including both intra- and inter-word patterns from the style reference, while ensuring the readability of text-lines that typically contain numerous characters.

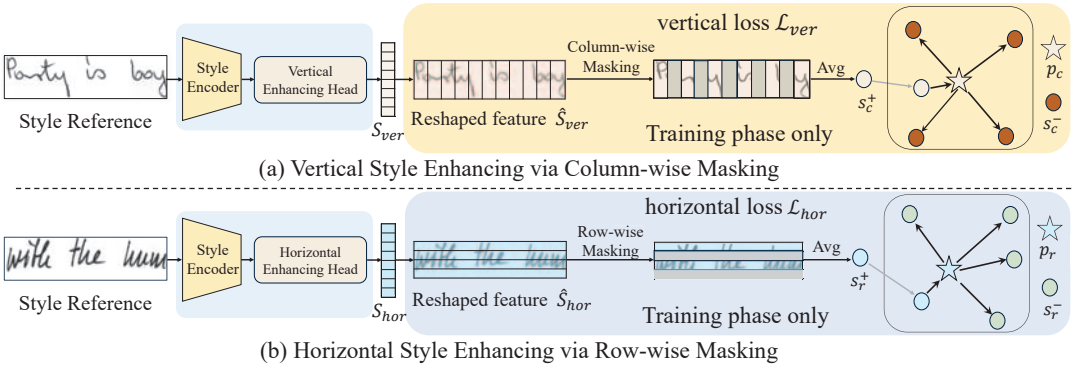
Conditional diffusion model. The diffusion model [17] generates realistic images by progressively denoising a random Gaussian noise input. To achieve controllable generation, the conditional diffusion model [4, 49, 62–64, 66] incorporates a condition signal c to guide the denoising process. Starting from pure Gaussian noise $x_T \sim \mathcal{N}(0, \mathcal{I})$, a denoising network p_θ iteratively refines the image over multiple timesteps to produce the target image x_0. The network p_θ, typically based on a U-Net architecture [50], integrates c via cross-attention or adaptive modulation layers. The training objective minimizes the mean squared error (MSE) between the predicted and real images: $\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_t, c}\left[\|x_0 - x_{\text{real}}\|^2\right]$. By leveraging condition signals such as text prompts and reference images, conditional diffusion models enable fine-grained control over the generation.

4. Method

4.1. Overall Scheme

To generate handwritten text-lines with enhanced content accuracy and style fidelity, we propose DiffBrush, a novel conditional diffusion generation method. As shown in Figure 3, the architecture of DiffBrush consists of three main components: content-decoupled style module, conditional diffusion generator, and multi-scale content discriminators.

The content-decoupled style module ξ_{style} aims to better capture the text-line styles of exemplar writers. To achieve this, we introduce a content-decoupled style learning strategy (cf. Section 4.2), which leverages two novel content-masking techniques and a style learning loss $\mathcal{L}_{style}$ to enhance text-line style modeling (cf. Figure 3). The extracted style features are then fused with content features from a content encoder to form the condition vector c within a blender module. Both the content encoder and blender module are designed based on One-DM [7], with further extensions (cf. Appendix B for details). Guided by c, the condi-



(a) Vertical Style Enhancing via Column-wise Masking

(b) Horizontal Style Enhancing via Row-wise Masking

Figure 4. Style learning via column- and row-wise masking. (a) An improved vertical style feature S_{ver} is extracted by the vertical enhancing head, guided by a vertical loss $\mathcal{L}_{ver}$. Specifically, during training, we begin reshaping S_{ver} into spatial feature $\hat{S}_{ver}$ and perform column-wise masking on $\hat{S}_{ver}$. After average pooling, s_c^+ is drawn closer to its corresponding writer proxy p_c, while the negatives s_c^- belonging to different writers are pushed away by p_c. (b) Similarly, a style feature S_{hor} is extracted by the horizontal enhancing head. We reshape S_{hor} and conduct row-wise masking. After pooling, s_r^+ is linked to its writer proxy p_r, and negatives s_r^- are pushed away.

tional diffusion generator $\mathcal{G}$ performs denoising process to synthesize realistic handwritten text-line image x_0.

However, training the generator $\mathcal{G}$ with solely the diffusion loss $\mathcal{L}_{diff}$ is insufficient to ensure the content readability of generated text lines. To address this, we introduce a multi-scale content learning strategy (cf. Section 4.3). Specifically, we develop a multi-scale discriminator $\mathcal{D}$ to evaluate the content correctness at both the line and word levels, thus providing more fine-grained content supervision $\mathcal{L}_{conent}$ for content adversarial learning between $\mathcal{G}$ and $\mathcal{D}$.

To summarize, the overall training objectives of our DiffBrush combines all three loss functions:

$$\mathcal{L}_{\mathcal{G}} = \mathcal{L}_{diff} + \mathcal{L}_{style} + \lambda \mathcal{L}_{content}, \tag{1}$$

where λ serves as a trade-off factor, and we empirically set it to 0.05 in training.

4.2. Content-decoupled Style Learning

As discussed in Section 1, text-line style learning is hindered by content interference, leading to ineffective style extraction. To address this, we propose to disrupt content information (by masking) to eliminate content interference during style learning, forcing the model to focus on essential style patterns. Specifically, as shown in Figure 3, style samples are first processed by a CNN-Transformer style encoder to obtain an initial style feature S with rich calligraphic attributes. To refine style representation, we introduce two dedicated style-enhancing heads, each incorporating a standard self-attention layer to extract fine-grained style features: S_{ver} (vertical) and S_{hor} (horizontal). These features are learned by using row- and column-wise masking for content disruption. The overall content-decoupled style loss $\mathcal{L}_{style}$ is formulated as the sum of a vertical enhancing loss $\mathcal{L}_{ver}$ and a horizontal enhancing loss $\mathcal{L}_{hor}$:

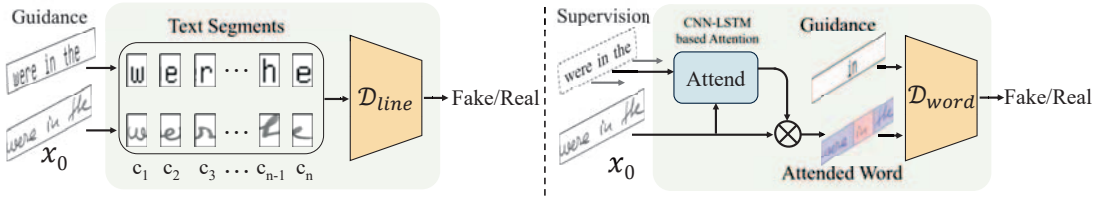
$$\mathcal{L}_{style} = \mathcal{L}_{ver} + \mathcal{L}_{hor}. \tag{2}$$

Vertical style enhancing via column-wise masking. The vertical enhancing head aims to enhance style learning in the vertical direction (*e.g.*, vertical alignment patterns) via column-wise masking. Specifically, as shown in Figure 4(a), we perform masking on S_{ver} by first reshaping the sequential feature S_{hor} back into spatial feature $\hat{S}_{ver} \in \mathbb{R}^{h \times w \times c}$. We then divide $\hat{S}_{ver}$ into several columns and randomly mask a subset of columns with equal probability to obtain an average feature $s_c \in \mathbb{R}^{h \times n \times c}$, where $n = w \cdot \rho$ and ρ is the masking ratio. After column-wise masking, we adopt a Proxy-NCA loss [25, 38] for style learning, which enforces style consistency within writers while distinguishing different writers. Specifically, our vertical enhancing loss $\mathcal{L}_{ver}$ assigns a proxy to each writer, treating it as an anchor to cluster the masked style features of the same writer while pushing apart those of different writers:

$$\mathcal{L}_{ver} = \frac{1}{|P_c^+|} \sum_{p_c \in P_c^+} \log\left(1 + \sum_{s_c \in S_c^+} e^{-f_c^+}\right) + \frac{1}{|P_c|} \sum_{p_c \in P_c} \log\left(1 + \sum_{s_c \in S_c^-} e^{f_c^-}\right), \tag{3}$$

where $S_c = \{s_c^i\}_{i=1}^N$ is a batch of masked style features, P_c denotes the set of proxies of all writers, and P_c^+ refers to the set of writers present in the current batch. For each proxy p_c, S_c is divided into a positive set S_c^+, consisting of s_c from the same writer as p_c, and a negative set $S_c^- = S_c - S_c^+$. The similarity between positive pairs is $f_c^+ = \alpha(g(s_c, p_c) - \delta)$ for $s_c \in S_c^+$, and that of negative pairs is $f_c^- = \alpha(g(s_c, p_c) + \delta)$ for $s_c \in S_c^-$, where $g(\cdot)$ is the cosine similarity, $\delta > 0$ is a margin and α is a scaling factor.

Horizontal style enhancing via row-wise masking. The horizontal enhancing head aims to enhance style learning in the horizontal direction (*e.g.*, word and character spacing) via row-wise masking. As shown in Figure 4(b), the masking operation is conducted in a similar way as column-wise making. Specifically, we reshape the sequential feature S_{hor} back into a spatial feature $\hat{S}_{hor} \in \mathbb{R}^{h \times w \times c}$ and then conduct random row-wise masking to obtain $s_r \in$



(a) Line Content Discriminator (b) Word Content Discriminator

Figure 5. Illustration of the multi-scale content discriminators. (a) For the line content discriminator $\mathcal{D}_{line}$, we concatenate the generated text-line image x_0 and guidance image along the channel dimension, slice the result into n segments, and then process them with a 3D CNN [55] to integrate context information. Finally, the line discriminator $\mathcal{D}_{line}$ evaluates each segment as real or fake. (b) For the word content discriminator $\mathcal{D}_{word}$, we utilize an attention module [52] to attend to individual words within the generated text line image x_0, and then input these words, along with corresponding content guidance, into D_{word} for realism discrimination.

$\mathbb{R}^{m\times w\times c}$, where $m = h \cdot \rho$. Our horizontal enhancing loss $\mathcal{L}_{hor}$ pulls the masked style features of the same writer together while pushing those of different writers apart:

$$\mathcal{L}_{hor} = \frac{1}{|P_r^+|}\sum_{p_r\in P_r^+}\log\left(1+\sum_{s_r\in S_r^+}e^{-f_r^+}\right)+\frac{1}{|P_r|}\sum_{p_r\in P_r}\log\left(1+\sum_{s_r\in S_r^-}e^{f_r^-}\right), \quad (4)$$

where we assign a proxy p_r to each writer and link it with all masked results S_r. The similarity between positive pairs is $f_r^+ = \alpha(g(s_r, p_r) - \delta)$ for $s_r \in S_r^+$, and that of negative pairs is $f_r^- = \alpha(g(s_r, p_r) + \delta)$ for $s_r \in S_r^-$.

4.3. Multi-Scale Content Learning

Existing methods [7, 9, 12, 23, 44] rely on content recognition losses to enhance the content readability of generated handwriting images. However, these losses, applied at the text-line level, prioritize global correctness but often fail to ensure character-level accuracy. For instance, in datasets like IAM [36], where a single text line averages 42 characters—approximately six times the length of a typical word, maintaining content integrity across the entire generated text becomes challenging. To address this problem, we introduce a multi-scale content learning strategy that provides finer-grained content supervision at both global (line) and local (word) levels for content adversarial training. As shown in Figure 5, the line content discriminator $\mathcal{D}_{line}$ evaluates the overall character order with a line discrimination loss $\mathcal{L}_{line}$, while the word content discriminator $\mathcal{D}_{word}$ ensures character-level accuracy through a word discrimination loss $\mathcal{L}_{word}$. The overall multi-scale content loss is then formulated as:

$$\mathcal{L}_{content} = \mathcal{L}_{line} + \mathcal{L}_{word}. \quad (5)$$

Line content discriminator. As shown in Figure 5(a), given the generated image x_0 after diffusion and the content guidance I_{line} without style information, the line discriminator $\mathcal{D}_{line}$ aims to determine whether the overall character order in x_0 matches that in I_{line}. Firstly, we set the wider of the I_{line} and x_0 as the benchmark, padding the narrower one with white background pixels for width alignment. We then concatenate x_0 and I_{line} along the channel dimension, and then slice the concatenated result into n non-overlapping segments $\{c_i\}_{i=1}^n$ from left to right. Afterwards, a 3D CNN [55] based discriminator $\mathcal{D}_{line}$ processes $\{c_i\}_{i=1}^n$ to incorporate global context information of characters, and determines whether this output is real or fake, providing feedback for the overall character order. The line discriminator loss $\mathcal{L}_{line}$ is formulated as:

$$\mathcal{L}_{line} = \log(\mathcal{D}_{line}(I_{line}, x_{real})) + \log(1 - \mathcal{D}_{line}(I_{line}, x_0)). \quad (6)$$

Word content discriminator. Compared to the line discriminator $\mathcal{D}_{line}$, the word discriminator $\mathcal{D}_{word}$ is designed to ensure that the text structure is correctly generated at the word level. However, accurately locating word positions within a whole text-line x_0 is non-trivial. Motivated by ASTER [52], we utilize an attention module with a CNN-LSTM architecture to obtain word positions.

As shown in Figure 5(b), given the generated image x_0 after diffusion, a CNN encoder first extracts spatial features $F_{map} \in \mathbb{R}^{h\times w\times c}$ from x_0, which is flattened into sequential features $H \in \mathbb{R}^{l\times c}$, where $l = h \times w$. The LSTM decoder then takes x_0 and a start-of-sequence (SOS) token as input, sequentially outputting attention maps for character positions until the end-of-sequence (EOS) token is reached. The character-level attention maps are then concatenated into word-level attention maps $A = \{a_t\}_{t=1}^T$ (cf. Figure 12 in Appendix), where $a_t \in \mathbb{R}^{h\times w}$ and T denotes the number of words in the text-line. Based on the attention maps, we extract attended words $\{x_{word}^t\}_{t=1}^T$, with $x_{word}^t = a_t \cdot x_0$. Lastly, each x_{word} and its corresponding content guidance I_{word} are fed into $\mathcal{D}_{word}$ for discrimination, which provides word-level content feedback for the generator to refine the local content readability. Specifically, the word discriminator loss $\mathcal{L}_{word}$ is formulated as:

$$\mathcal{L}_{word} = \sum_{i=1}^{T}\log(\mathcal{D}_{word}(I_{word}^i, x_{real}^i)) + \sum_{i=1}^{T}\log(1 - \mathcal{D}_{word}(I_{word}^i, x_{word}^i)), \quad (7)$$

where i represents the i-th word in a text line.

Datasets	Method	Shot	HWD ↓	D_{CER} ↓	D_{WER} ↓	FID ↓	IS ↑	GS ↓
IAM	TS-GAN [9]	one	2.11	44.20	87.13	16.76	1.76	2.87×10^{-2}
	CSA-GAN [23]	few	2.25	42.27	84.14	13.52	1.74	1.62×10^{-2}
	VATr [44]	few	1.87	28.80	71.77	12.51	1.69	1.45×10^{-2}
	DiffusionPen [40]	few	1.72	54.75	84.70	10.24	1.83	6.42×10^{-3}
	One-DM [7]	one	1.80	20.91	54.27	10.60	1.82	8.42×10^{-3}
	Ours	one	**1.41**	**8.59**	**28.60**	**8.69**	**1.85**	$\mathbf{2.35 \times 10^{-3}}$
CVL	CSA-GAN [23]	few	1.72	41.64	72.02	8.71	1.48	6.71×10^{-2}
	VATr [44]	few	1.50	38.49	66.33	9.04	1.44	1.43×10^{-1}
	DiffusionPen [40]	few	1.32	55.94	88.36	11.90	1.59	5.08×10^{-2}
	One-DM [7]	one	1.47	32.42	63.35	11.95	1.46	1.29×10^{-1}
	Ours	one	**1.06**	**20.92**	**36.38**	**7.57**	**1.70**	$\mathbf{2.96 \times 10^{-2}}$

Table 1. Comparisons with baselines on handwritten text-line generation on IAM and CVL. All methods are trained on the same training set and evaluated using the same protocols. The "Shot" column indicates the number of text-line style references required for each method.

5. Experiments

5.1. Experimental Settings

Evaluation dataset. To evaluate our DiffBrush in generating handwritten text-line, we use the widely adopted handwriting datasets IAM [36] and CVL [26]. IAM contains 13,353 English text-line images belonging to 657 unique writers. Following the protocol of CSA-GAN [23], we use text-lines from 496 writers for training and the remaining 161 writers for testing. CVL dataset consists of handwritten text-lines from 310 writers in both English and German. For our experiments, we use the English portion, consisting of 11,007 text-lines, and follow the standard CVL split, with 283 writers for training and 27 for testing. In all experiments, we resize the images to a height of 64 pixels while preserving their aspect ratio, as done in previous works[7, 9, 23]. To manage varying widths, images with a width smaller than 1024 pixels are padded, whereas those exceeding 1024 are resized to a fixed size of 64×1024. We also evaluate DiffBrush on popular Chinese dataset CASIA-HWDB (2.0–2.2) [32] (cf. Appendix E for more details.)

Evaluation metrics. 1) We use the newly proposed Handwriting Distance (HWD) [45], specifically designed for handwriting style evaluation. HWD computes the Euclidean distance between features extracted by a VGG16 network pre-trained on a large corpus of handwritten text images. 2) For content evaluation, we follow latest studies [7, 40, 41] by using the generated training sets from each method to train an OCR system [48] and report its recognition performance on the real test set in terms of CER and WER. We name these new evaluation metrics D_{CER} and D_{WER}, with further discussions provided in Appendix C. 3) We use Fréchet Inception Distance (FID) [15], Inception Score (IS) [51], and Geometry Score (GS) [24] to measure the visual quality of generated images. 4) We also conduct user studies to quantify the subjective quality of the generated handwritten text-line images in Appendix D.

Implementation details. In all experiments, we use a randomly selected text-line sample as the style reference. In DiffBrush, both the style and content encoders are based on a ResNet18, followed by 2 transformer encoder layers. Line discriminator uses three 3D convolution layers, and word discriminator has three 2D convolution layers. The model is trained for 800 epochs on eight RTX 4090 GPUs using the AdamW optimizer with a learning rate of 10^{-4}. We set masking ratio ρ to 0.5 after a grid search (cf. Table 5 in Appendix). We randomly drop the condition c with the probability 0.1 for classifier-free training [16]. During inference, we adopt a classifier-free guidance scale of 0.2 and use DDIM [53] with 50 steps to accelerate the process. More details are put in Appendix B.

Compared Methods. We compare DiffBrush with state-of-the-art handwritten text-line generation methods, including TS-GAN [9], CSA-GAN [23], and advanced word-level handwritten text generation approaches like VATr [44], DiffusionPen [40] and One-DM [7]. We provide implementation details of word-level methods in Appendix G.

5.2. Main Results

Styled handwritten text-line generation. We assess DiffBrush for generating handwritten text-line images with desired style and specific content. To quantify style similarity, following CSA-GAN [23], we generate text-line images for each method using style information from test set and content input from a subset of WikiText-103 [37]. We then calculate the HWD between the generated and real samples for each writer, and finally average the results.

The quantitative results in Table 1 show that DiffBrush outperforms all state-of-the-art methods on both IAM and CVL datasets. Specifically, it improves HWD by 18.02% ($1.72 \rightarrow 1.41$) on IAM and 19.69% ($1.32 \rightarrow 1.06$) on CVL compared to the second-best method, highlighting its superior style imitation ability. Moreover, DiffBrush achieves significantly lower D_{CER} and D_{WER} on both IAM and CVL datasets, demonstrating its advantage in content readability. In contrast, DiffusionPen [40] yields the highest D_{CER} and D_{WER} due to ineffective content supervision.

We provide qualitative results to intuitively explain the benefit of our DiffBrush in Figure 6. TS-GAN struggles to accurately capture the style patterns of reference sam-

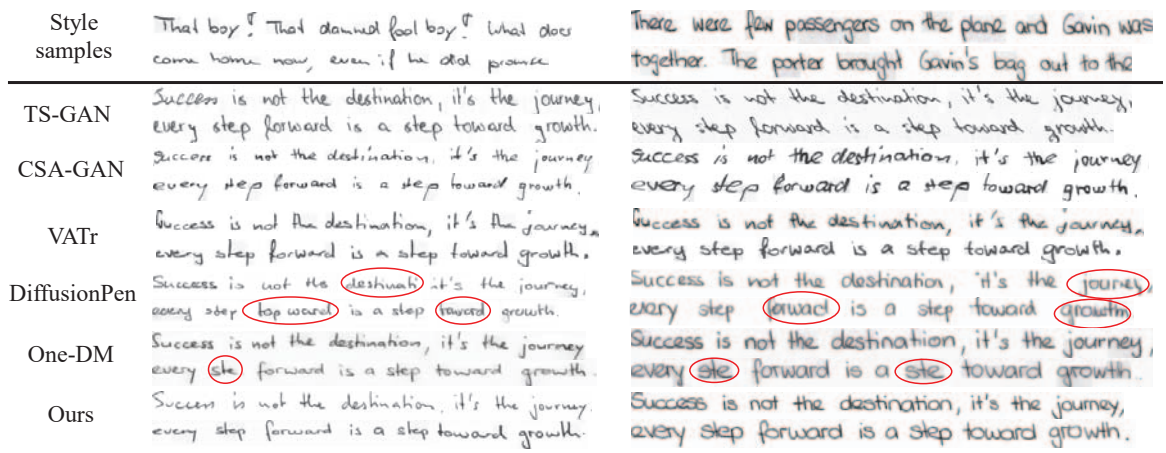


Figure 6. Qualitative comparisons between our method and state-of-the-art approaches for handwritten text-line generation, conditioned on out-of-vocabulary (OOV) textual content and unseen styles from the IAM test dataset. We use the same guiding text, "Success is not the destination, it's the journey, every step forward is a step toward growth." for all methods, instructing them to generate the text in different handwriting styles. The red circles highlight missing characters or structural errors. Better zoom in 200%.

Style sample	she had been sufficiently	HWD↓	D_{CER}↓	D_{WER} ↓
Base+ε_{single}+without masking	gave a long cous cough cough	1.82	56.02	87.25
Base+ε_{single}+random masking	gave a cough a long rough	1.75	55.41	86.42
Base+ξ_{style}	gave a bugh cough raucous	1.47	54.64	84.33
Base+ξ_{style}+$\mathcal{D}_{word}$	gave a long raucous cough cough	1.42	14.61	43.93
Base+ξ_{style}+$\mathcal{D}_{line}$	gave a long rcous cough	1.44	15.28	43.31
Base+ξ_{style}+$\mathcal{D}_{line}$+$\mathcal{D}_{word}$	gave a long raucous cough	**1.41**	**8.59**	**28.60**

Figure 7. Ablation studies of the content-decoupled style module (ξ_{style}) and the multi-scale content discriminators (*i.e.,* $\mathcal{D}_{line}$ and $\mathcal{D}_{word}$) based on IAM test set. ε_{single} denotes a single CNN-Transformer style encoder (ResNet18 followed by 3 transformer encoder layers). The red boxes highlight failure instances of structure preservation, whereas the blue box points out an incorrect repetitive word.

ples, like ink color and stroke width. CSA-GAN produces samples that lack style consistency, including inconsistent character slant, ink color, and stroke width. VATr has difficulty maintaining vertical alignment between words in the synthesized text lines. DiffusionPen struggles to ensure the content readability of generation results. One-DM occasionally generates text lines with missing or incorrect characters. Conversely, our DiffBrush excels at generating precise character details while maintaining overall consistency.

Style-agnostic text-line generation. We further evaluate DiffBrush's ability to generate realistic handwritten text-line images, independent of style imitation. Following TS-GAN [9], each method generates 25k random text-line images to calculate FID against all training samples, and 5k random samples for GS calculation, compared with 5k samples from the test set. Besides, we generate the entire test set using each method and evaluate the results using the IS. As shown in Table 1, DiffBrush achieves the highest performance across FID, IS, and GS metrics on both IAM and CVL datasets, further demonstrating its ability to generate superior-quality handwritten text-line images.

5.3. Analysis

In this section, we conduct ablation studies to analyze our DiffBrush. We provide more analyses in Appendix, including generalization evaluation on various style backgrounds, **failure case analysis**, ablation results on masking ratio, enriching datasets to train recognizer, style interpolation results, style evaluation results in terms of WIER [12, 61], discussions about fine-grained style learning.

Quantitative evaluation of style module and content discriminators. We perform multiple ablation studies to analyze different components. Quantitative results in Figure 7 reveal that: (1) Compared to two variants—a basic style encoder without masking, and with random masking (best masking ratio 0.5)—our style module ξ_{style} significantly improves HWD by 19.23% (1.82 $\rightarrow$ 1.47), and 16.00% (1.75 $\rightarrow$ 1.47), respectively. This highlights the effectiveness of ξ_{style} in style learning. (2) The combination of the $\mathcal{D}_{word}$ and $\mathcal{D}_{line}$ leads to significant improvements in terms of D_{CER} and D_{WER} without reducing HWD. This is achieved by employing style-free conditional discriminators, which

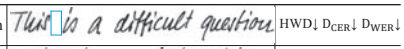
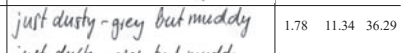
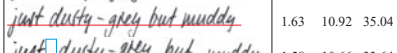
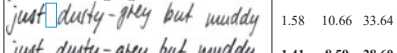
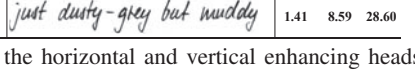
Style module in DiffBrush		HWD↓	D_{CER}↓	D_{WER}↓
Single style encoder		1.78	11.34	36.29
Ver. enhancing head only		1.63	10.92	35.04
Hor. enhancing head only		1.58	10.66	33.64
Our style module		**1.41**	**8.59**	**28.60**

Figure 8. Effect of the horizontal and vertical enhancing heads on IAM test set. Red lines highlight vertical alignment of words, while blue boxes denote the word spacing. "Single style encoder" is a basic CNN-Transformer encoder without masking.

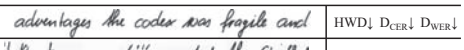
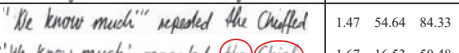
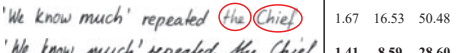
Style sample		HWD↓	D_{CER}↓	D_{WER}↓
Base+ξ_{style}		1.47	54.64	84.33
Base+ξ_{style}+CTC		1.67	16.53	50.48
Base+ξ_{style} +$\mathcal{D}$		**1.41**	**8.59**	**28.60**

Figure 9. Effect of discriminators $\mathcal{D}$ and CTC recognizer [14] under their best trade-off factor. Red circles indicate handwritten texts with simplified style (*i.e.*, regular texts with standard strokes).

focus solely on forcing the generator to enhance content readability, without impeding the learning of style.

Qualitative evaluation of style module and content discriminators. We conduct visual ablation experiments to further analyze each module in our DiffBrush. As shown in Figure 7, we observe that the first two basic baselines show clear drawbacks in both style imitation and content readability. Adding our style module significantly improves style reproduction, such as ink color and stroke width, but content readability remains poor. Adding $\mathcal{D}_{word}$ alone improves the character details. However, it struggles to maintain overall content readability, leading to issues like word repetitions and shifts. Employing $\mathcal{D}_{line}$ solely enhances overall content readability, but character detail issues still remain. In contrast, the best generation results are achieved when both line- and word-level discriminators are used.

Discussions about style learning. We conduct ablation study on the style module to analyze the differences between two style-enhancing heads. As shown in Figure 8, adding either the vertical or horizontal enhancing head improves text-line style quality in terms of HWD. The vertical head enhances the style imitation ability, particularly in maintaining consistent vertical word alignment. Meanwhile, the horizontal head also improves the style learning, like horizontal spacing patterns. These findings support our motivation that content-masking strategies in different directions help the effective style learning (cf. Figure 2). Finally, it is worth emphasizing that our style features contain complete style information as they are extracted from the entire style reference before any masking (cf. Figure 3).

Discussions on discriminators and CTC. The quantitative results in Figure 9 show that incorporating CTC recognizer [14] significantly reduces D_{CER} and D_{WER}, while also impairing the style evaluation (HWD). Visualization

Style sample		HWD↓	D_{CER}↓	D_{WER}↓	FID↓
VATr+assembling		2.46	36.58	80.94	29.87
One-DM+assembling		2.37	26.50	62.79	26.15
DiffusionPen+assembling		2.19	28.64	71.25	25.24
DiffBrush (Ours)		**1.41**	**8.59**	**28.60**	**8.69**

Figure 10. Comparisons between directly generated text-lines and text-lines assembled by isolated words on IAM test set.

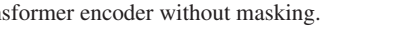
Method		HWD↓	D_{AR}↑	D_{CR}↑	FID↓
One-DM		1.09	81.99	82.80	17.38
Ours		**0.73**	**96.24**	**96.65**	**7.87**

Figure 11. Quantitative and qualitative comparisons with One-DM [7] on Chinese handwritten text-line generation. The blue boxes highlight the character spacing, while the red circles emphasize the incorrect character structures.

results in Figure 9 intuitively explain the reasons for this degradation. The CTC version tends to produce texts with simplified styles that differ significantly from style samples. Conversely, our discriminators enhance content readability while preserving style mimicry performance. We provide more experiment details and visual results in Appendix E.

Discussions on directly generated and assembled text-lines. We conduct experiments to demonstrate the superiority of directly generated text lines over those obtained through concatenation. To this end, we apply the assembling strategy in DiffusionPen [40] to enable the official word-level generation baselines to generate text-line image (cf. Appendix G). The results in Figure 10 demonstrate the superiority of our method over those non-text line methods. More experimental results are provided in Appendix H.

Applications to Chinese text-line generation. We assess DiffBrush's ability to generate Chinese scripts, a challenging task due to thousands of character categories and their complex structures. Both quantitative and qualitative results are provided on Figure 11. We observe our DiffBrush effectively handles Chinese handwritten text-lines in terms of style imitation and content fidelity. More experimental details and visualizations are put in Appendix E.

6. Conclusion

In this paper, we introduce DiffBrush, a novel diffusion model tailored for handwritten text-line generation. To the best of our knowledge, this is among the first exploration of diffusion models for this task. For better style learning and content guidance, we propose a content-decoupled style learning strategy that significantly enhances style imitation and multi-scale content discriminators that supervise textual content at both the line and word levels while preserving style imitation performance. Promising results on three widely-used handwritten datasets verify the effectiveness of our DiffBrush. In the future, we plan to explore its potential to support other generative tasks, such as font generation.

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